

Hong Kong Kindergarten Accessibility Analysis

<https://arcg.is/04OXnm>

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1 Section 1: The Setting

1.1 Status Quo

Hong Kong is currently home to more than 1,220 kindergartens spread across its 18 administrative districts, resulting in a fragmented accessibility landscape that appears comprehensive at first glance. The present spatial pattern reveals a pronounced clustering correlation between population density and kindergarten density in the core urban zones, including Hong Kong Island, the Kowloon Peninsula, and certain new towns.

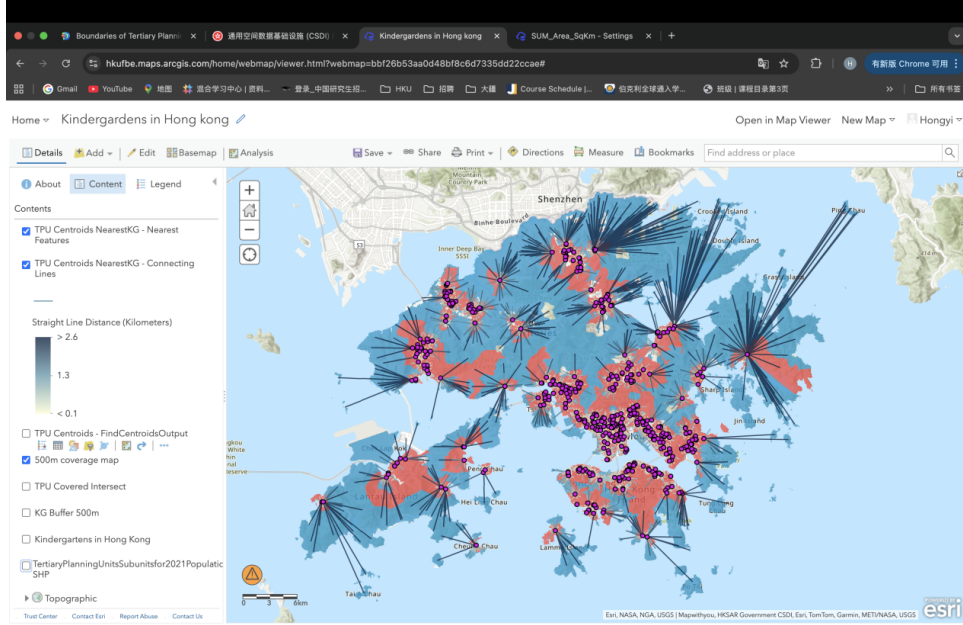


Figure 1: Spatial clustering of kindergartens and population density

1.2 Current State

Most urban TPUs lie within a 500-meter walking radius of at least one kindergarten, and densely urbanized neighborhoods benefit from both a high concentration of such facilities and short access distances. This aggregated perspective, however, obscures significant inherent fragmentation in provision. While the urban core boasts adequate kindergarten coverage, the peripheral New Territories, mountainous areas, and outlying islands face either a complete absence of nearby kindergarten services or substantially greater travel time burdens.

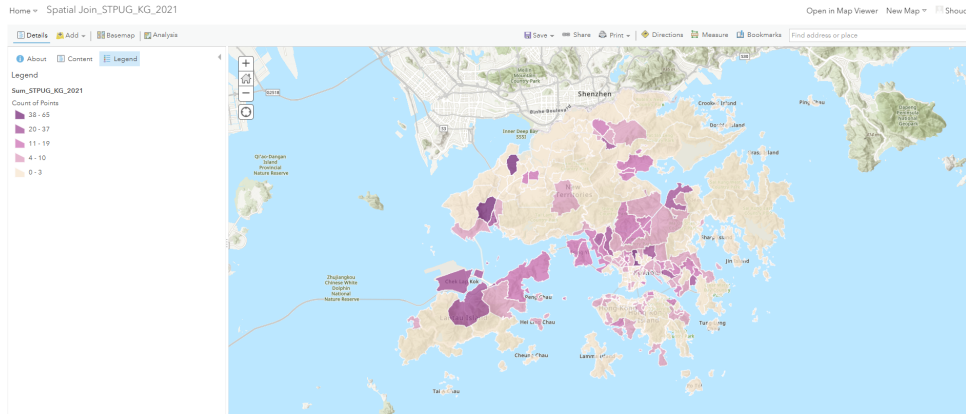


Figure 2: Accessibility gaps in peripheral regions

2 Section 2: The Hook

2.1 The Hook

Although urban centers appear to have adequate kindergarten coverage, Hong Kong is in fact experiencing a crisis in early childhood education defined not by mere accessibility gaps, but by severe systemic capacity shortages. This dilemma manifests as a striking paradox: while kindergartens are present in most districts, the demand burden on individual facilities is critically imbalanced. In the majority of neighborhoods, each kindergarten is responsible for serving over 8,430 children aged 3–5, creating unsustainable operational pressure even in areas nominally covered by services.

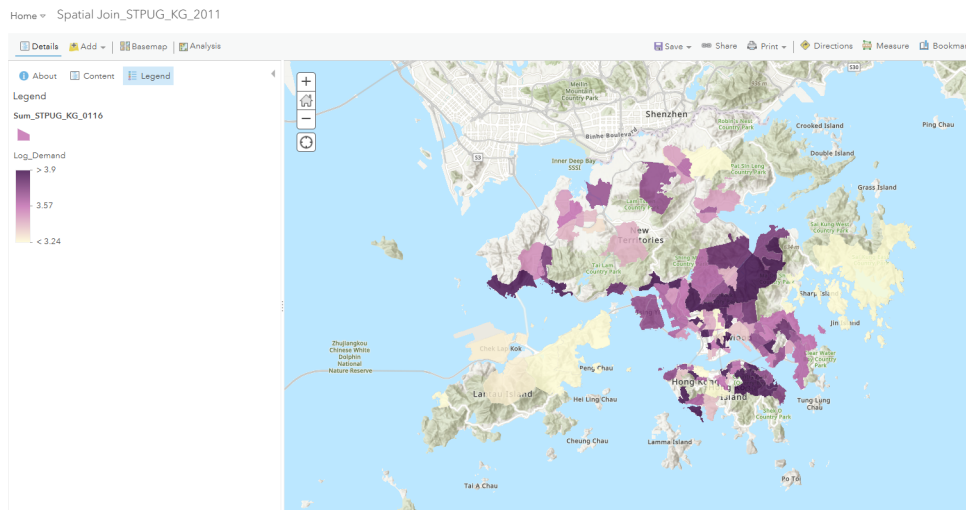


Figure 3: Capacity pressure per kindergarten

2.2 The Conflict Over Time

This crisis is intensifying over time. A longitudinal analysis spanning 2011–2021 reveals that shortages have worsened markedly in key regions such as Chek Lap Kok, central Lantau Island, and parts of Hong Kong Island. While some outlying islands with declining child populations show minor improvements, the overall trend remains concerning.

Even in Hong Kong’s best-performing districts—such as northeastern Lantau Island and Sai Kung—the provision remains low, with only 2 to 3.5 kindergartens available per 1,000 children. This ratio underscores systemic capacity constraints rather than isolated or localized gaps.

These patterns indicate that kindergarten resource allocation suffers not only from spatial inequity but also from persistent structural strain over time, necessitating dynamic planning and sustained policy intervention to ensure long-term adequacy and equity in early childhood education provision.

3 Section 3: Rising Insights

3.1 Demographic and Supply Fundamentals

With approximately 153,562 children aged 3-5 distributed unevenly across the territory, demand density varies dramatically by district. Sha Tin (14,213 children), Yuen Long (13,064), and Kwun Tong (12,475) carry the heaviest loads, while Wan Chai (3,592) and Islands District (3,782) show smaller populations. However, supply does not track linearly with demand. Supply-demand ratios range from 0.33 kindergartens per 100 children in Yau Tsim Mong to 1.61 in Islands District.

3.2 Regression Evidence

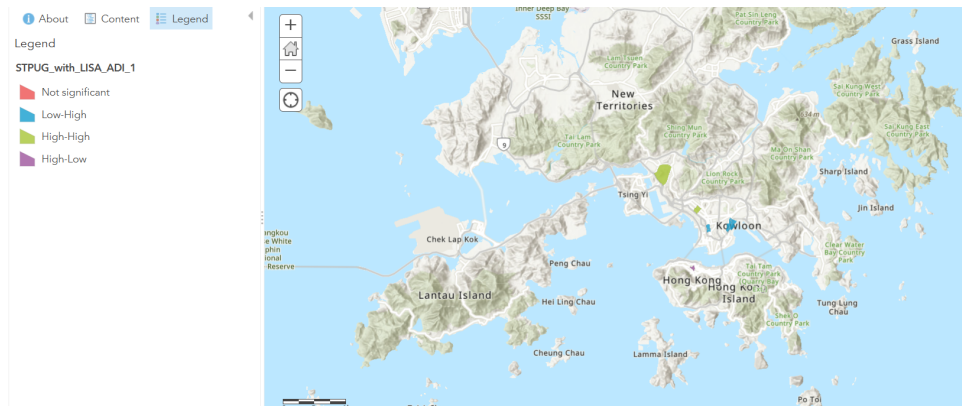
Controlling for demand fundamentals through regression analysis (Kindergarten Density vs. Child Density) reveals that child density explains 75.54% ($R^2 = 0.7554$) of kindergarten density variation—indicating that demand is generally the primary driver of supply location. Yet significant systematic residuals persist:

- **Systematic Oversupply:** Sham Shui Po District (standardized residual: 2.34) and Kowloon City District (2.17) exhibit kindergarten densities significantly exceeding model predictions based on child populations.
- **Systematic Undersupply Risk:** Yau Tsim Mong District (standardized residual: -1.90) falls just below the statistical threshold for significant undersupply, with actual density at 1.64 kindergartens/km² against an expected 3.89 kindergartens/km².

3.3 Spatial Clustering Analysis

Local Indicators of Spatial Association (LISA) analysis on the Accessibility Difference Index (ADI) at the Street Block Planning Unit Group (STPUG) level reveals that these mismatches are localized phenomena rather than city-wide patterns. Only 7 out of 211 STPUGs show statistically significant clustering ($p < 0.05$), indicating that oversupply and undersupply exist as fragmented “hotspots” rather than continuous regional failures. The pattern is characterized by:

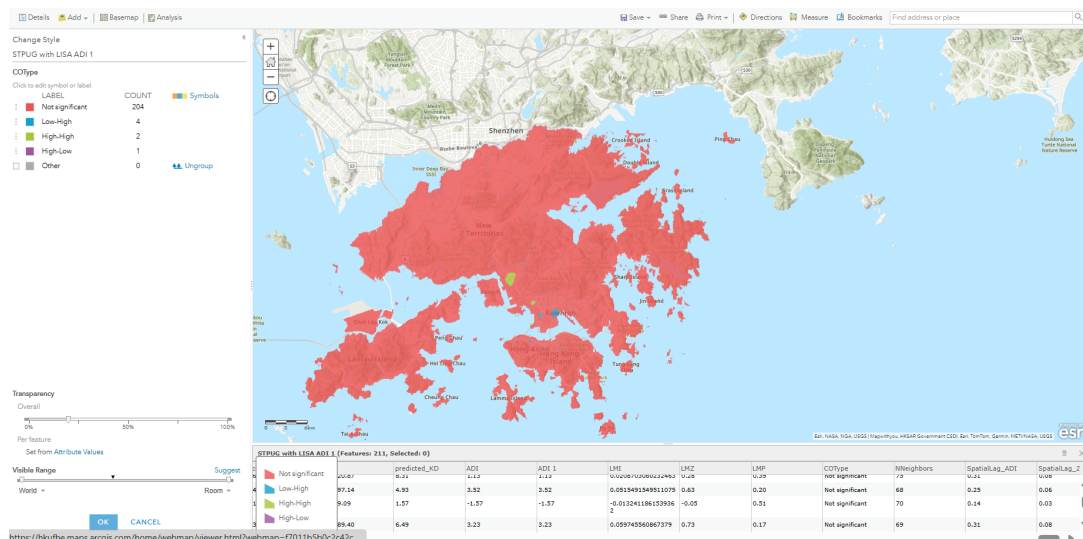
- High-High clusters (2 STPUGs): Localized oversupply pockets
- Low-High outliers (4 STPUGs): Undersupply pockets within oversupplied neighborhoods (priority intervention zones)
- High-Low outlier (1 STPUG): Sharp local contrast requiring investigation



4 Section 4: Aha Moment

4.1 The Central Insight

Hong Kong's kindergarten crisis is not a uniform shortage but a "fragmented mismatch" characterized by the coexistence of extreme oversupply and critical undersupply within close proximity. The market exhibits spatial inefficiency: while 15 of 18 districts show balanced supply-demand relationships, resources are effectively "trapped" in oversupplied core Kowloon districts (Sham Shui Po, Kowloon City) due to historical development patterns, while emerging demand centers and commercial hubs face acute shortages.



4.2 Recommendation for New Location

4.2.1 Immediate Action Zones (New Construction Priority)

1. **Service Gap Areas:** Districts with Point_Count = 0 (no kindergarten service) require immediate infrastructure investment to eliminate absolute access deserts.
2. **Extreme Shortage Zones:** Areas where KG_Per_1000Child falls between 0 and 0.5 (indicating > 2,000 children competing per kindergarten), specifically consistent

shortage zones identified across both 2011 and 2021 datasets in Chek Lap Kok and central Lantau.

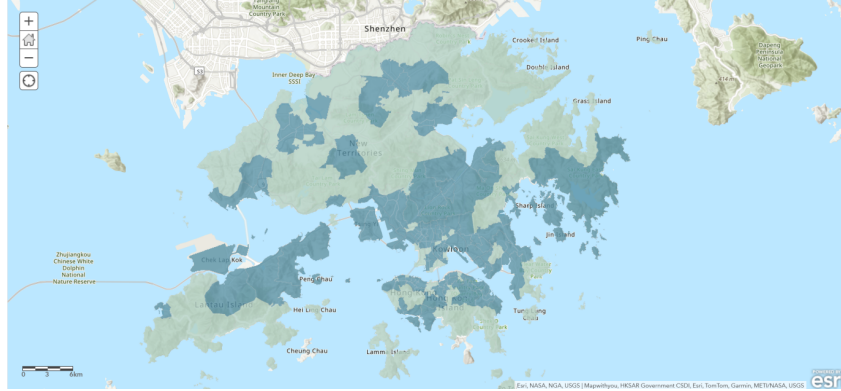


Figure 6: Extreme shortage zones

4.2.2 Resource Rebalancing Opportunities

Sham Shui Po and Kowloon City: These districts show systematic oversupply with densities of 6.70 and 6.66 kindergartens/km² respectively—2.5+ kindergartens/km² above predicted demand levels. Planning authorities should investigate whether these represent duplicate constructions or excess capacity that could be conceptually rebalanced to adjacent underserved areas.

4.2.3 Special Commercial District Intervention

Yau Tsim Mong District: As Hong Kong’s commercial and tourism center, this district faces unique barriers (high land costs, limited residential land) resulting in the territory’s lowest supply-demand ratio (0.33). This requires targeted policy intervention to overcome market-driven undersupply in high-cost urban cores.

4.2.4 Precautionary Planning

Chek Lap Kok & Central Lantau: These areas showed significant deterioration from 2011 to 2021, indicating rapidly escalating demand outstripping supply. Proactive capacity expansion is required before crisis thresholds are reached.

5 Section 5: Solution & Next Steps

Therefore, transition from district-level aggregate planning to STPUG-level spatial optimization, recognizing that supply-demand mismatches operate at highly localized scales are necessary. For the policy makers, it is also crucial to integrate the regression model ($R^2 = 0.7554$) into standard planning protocols, using predicted kindergarten density as a benchmark to flag districts that require intervention.

5.1 Expansion Sites ($\text{ADI} \geq 1.5$)

Target: 4 high-priority STPUGs exhibiting severe undersupply (Low-High spatial outliers) and 12 district-level zones with absolute service gaps.

Table 1: Expansion Sites ($ADI \geq 1.5$)

Location ID	ADI Score (Residual)	Spatial Type	Intervention Strategy
STPUG228	-14.58	Low-High outlier ($p = 0.019$)	Immediate new facility construction; target capacity: 2-3 kindergartens to absorb excess demand from surrounding oversupplied neighbours.
STPUG241	-9.18	Low-High outlier ($p = 0.019$)	Capacity expansion via modular units or session licensing until permanent construction is completed.
STPUG229	-9.87	Low-High outlier ($p = 0.045$)	Fast-track land-use conversion for early-childhood education; prioritise public-private partnership given high demand pressure.
STPUG242	-7.45	Low-High outlier ($p = 0.046$)	Temporary satellite campuses of nearby oversupplied districts until standalone facilities are built.
Chek Lap Kok / Central Lantau	< 0.5 kg / 1,000 children	Extreme short-age zone (2011-2021 deterioration)	Emergency procurement of 3-4 large-scale kindergartens to reverse the decade-long shortage trajectory.
Point.Count = 0 TPUs	No-access bi-nary	Absolute service deserts	Greenfield construction

5.2 Consolidation Sites ($ADI \leq 0.5$)

Target: 2 high-significance STPUGs and 2 administrative districts with systematic over-supply.

Table 2: Consolidation Sites ($ADI \leq 0.5$)

Location ID	ADI (Residual)	Score	Density Status	Reallocation Strategy
STPUG265	+15.25		High-High cluster ($p = 0.017$)	Flag for capacity audit; redirect 30-40% of planned facility maintenance budgets to expansion sites.
STPUG326	+5.53		High-High cluster ($p = 0.022$)	Consolidate redundant session slots; explore facility-sharing agreements with adjacent under-supplied zones.
Sham Shui Po District	+2.34 (Standard- ized Residual)		6.70 kg/km ² ac- tual vs. 4.02 kg/km ² predicted	Moratorium on new kindergarten licenses; reallocate planning permissions to Yau Tsim Mong.
Kowloon City District	+2.17 (Standard- ized Residual)		6.66 kg/km ² ac- tual vs. 4.11 kg/km ² predicted	Initiate "teaching resource mobility" programme—relocate excess teaching staff to Chek Lap Kok/Central Lantau expansion sites.

5.3 Quantified Impact Target

1. Spatial Inefficiency Reduction:

Eliminate all 4 Low-High outlier clusters (STPUGs228,229,241,242) through targeted expansion, and dissolve the 2 High-High clusters (STPUGs265,326) via reallocation.

2. Metric:

Reduce the number of statistically significant spatial autocorrelation clusters (Moran'sI) from 7 to ≤ 5 within 18 months, indicating a shift from clustered mismatch toward

random (equitable) distribution.

5.4 Strategic Implementation Roadmap (1-3 Years)

To ensure the cluster structure shifts sustainably over the next 5 years, the following mechanisms will operationalise the ADI thresholds:

1. Dynamic ADI-Triggered Acquisition Protocol

Deploy a real-time GIS dashboard that updates ADI scores quarterly using the regression model ($R^2 = 0.7554$; Kindergarten Density = $0.0638 + 0.007662 \times \text{Child Density}$).

- Trigger: When an STPUG's rolling 12-month ADI exceeds +1.5, automatic planning alerts trigger consolidation reviews.
- Trigger: When ADI falls below -1.5, pre-approved "rapid response" capital funds are released for site acquisition.

2. Cross-Boundary Resource Rebalancing

Establish a Kindergarten Resource Transfer Mechanism between Consolidation and Expansion zones:

- Years 1-2: Pilot transferable quotas between Sham Shui Po and Yau Tsim Mong, and between Kowloon City and the Chek Lap Kok/Lantau corridor.
- Years 2-3: Scale to territory-wide "ADI credits," allowing oversupplied districts to offset construction costs in undersupplied districts.

3. Predictive Cluster Prevention

Utilise the Supply-Demand Ratio model (current average = 0.80 kindergartens per 100 children) to project demand growth in high-fertility zones (e.g., Yuen Long, Sha Tin).

- Action: Pre-emptively secure land in TPUs forecast to cross $ADI \geq 1.5$ within 5 years.

4. Zoning Code Reformation

- Commercial District Adjustment: Amend land-use statutes in Yau Tsim Mong to mandate kindergarten provision in all new commercial-residential mixed developments $> 20,000 \text{ m}^2$ GFA.
- Oversupply Caps: Implement statutory maximum kindergarten densities (based on $ADI \leq 0.5$ thresholds) in Sham Shui Po and Kowloon City.

5.5 Long-Term System Evolution (3-5 Years): Cluster Transition Targets

By Year 5, the spatial pattern should shift from "fragmented mismatch" to "diffused equilibrium":

Table 3: Cluster Transition Targets (3-5 Years)

Current Cluster Type	5-Year State	Target	Mechanism
Low-High Outliers (4 zones)	Dissolved → (Non-significant)	Balanced	Expansion construction raises local ADI into the 0.5-1.5 equilibrium band.
High-High Clusters (2 zones)	De-clustered → Dispersed balanced supply		Consolidation reduces local density; facilities reallocated to adjacent districts.
High-Low Outlier (1 zone: STPUG122)	Investigated → Resolved boundary effect		Determine if ADI = +23.50 is a data artefact or genuine mismatch; rectify.

5.6 Success Metrics (2029 Targets)

1. Moran’s I: Reduce global spatial autocorrelation index by $\geq 25\%$.
2. Coverage: $ZeroTPUswithPoint_Count = 0$.
3. Efficiency: Reduce districts with $KG_Per_1000Child < 0.5$ from current levels to < 3 districts.
4. Equity: Standardised residuals for all 18 districts constrained within ± 1.5 .

Appendix

Detailed Statistical Tables

Table 4: Kindergarten and Child Quantity Statistics by District

District	KG Count	Children 3-5	S-D Ratio	Std. Res.	Status
Sham Shui Po District	85	6,482	1.31	2.34	Oversupply
Kowloon City District	113	9,029	1.25	2.17	Oversupply
Islands District	61	3,782	1.61	-0.02	Balanced
North District	72	6,702	1.07	0.03	Balanced
Yuen Long District	116	13,064	0.89	0.01	Balanced
Central and Western District	45	5,244	0.86	0.12	Balanced
Kwai Tsing District	97	10,810	0.90	0.16	Balanced
Kwun Tong District	101	12,475	0.81	0.23	Balanced
Tuen Mun District	85	10,545	0.81	-0.04	Balanced
Sai Kung District	76	9,674	0.79	-0.05	Balanced
Tai Po District	48	6,702	0.72	-0.06	Balanced
Sha Tin District	99	14,213	0.70	-0.15	Balanced
Southern District	31	6,008	0.52	-0.17	Balanced
Tsuen Wan District	31	6,851	0.45	-0.24	Balanced
Eastern District	64	12,680	0.50	-0.99	Balanced
Wong Tai Sin District	52	9,063	0.57	-1.43	Balanced
Yau Tsim Mong District	22	6,637	0.33	-1.90	Potential Imbalance
Wan Chai District	26	3,592	0.72	—	Data Missing

Abbreviations: KG: Kindergarten; S-D Ratio: Supply-Demand Ratio; Std. Res.: Standardized Residual.

Table 5: Spatial Analysis Units

Unit	Description	Feature Count	Source
TPU	Tertiary Planning Unit – polygon boundaries defining residential analysis zones	18 Districts/Sub-zones	Census and Statistics Department (2021)
STPUG	Street Block Planning Unit Group – fine-scale polygon units for granular supply-demand analysis	211 features	Census and Statistics Department (2021)
TPU Centroid	Geometric center of TPU polygon used for distance calculations	Point features	ArcGIS Feature to Point tool

Table 6: Basic Accessibility Indicators

Variable	Calculation / Definition	Data Source / GIS Tool
Geometric Variables		
area_sqkm	Polygon area of STPUG converted to square kilometres	ArcGIS Calculate Geometry
Count Variables		
Join_Count / Point_Count	Number of kindergarten points falling within each STPUG polygon	Spatial Join (Target: STPUG; Join: Kindergarten points)
Kindergarten_Count	Total number of kindergartens per administrative district (18 districts)	Aggregated from Point_Count
children_u6	Total children aged 0–5 per STPUG; sum of age_1 through age_5 census fields	2021 Hong Kong Population Census (STPUG level)
Children_3_5	Estimated children aged 3–5 per district (kindergarten-eligible population)	2021 Hong Kong Population Census (District level)
Density Variables		
KG_Density	Kindergarten Density = $\text{Join_Count} / \text{area_sqkm}$	Calculated Field
Child_Density / Children_Density	Number of children aged 3–5 per square kilometre = $\text{Children_3_5} / \text{area_sqkm}$	Derived from census and geometry data
Ratio Variables		
Demand_Ratio	Service load per facility = $\text{Children_3_5} / \text{Total Kindergartens in zone}$	Derived field indicating competition intensity

Table 7: Distance-Based Accessibility Indicators

Variable	Calculation / Definition	GIS Tool / Method
Nearest_Dist	Straight-line (Euclidean) distance from TPU centroid to the nearest kindergarten point location	ArcGIS "Find Nearest" / "Generate Near Table"
Coverage_500m	Indicator = 1 if kindergarten j such that $d(\text{Centroid}_i, \text{KG}_j) \leq 500m$; otherwise 0	Arcade Expression: if (\$feature.NEAR_DIST ≤ 500)return1else0
No_Access	Indicator = 1 if zero kindergartens exist within or adjacent to the TPU polygon (Point_Count == 0); otherwise 0	Spatial Join count validation

Table 8: Supply-Demand Balance Classification (KG_Per_1000Child)

KG_Per_1000Child Range	Interpretation	Supply Status
≥ 0 and ≤ 0.5	>2,000 children competing per kindergarten	Critical Shortage – Severe undersupply requiring immediate intervention
> 0.5 and ≤ 1.0	1,000–2,000 children per kindergarten	High Pressure – Inadequate capacity relative to demand
> 1.0 and ≤ 2.0	1,000 children per kindergarten	Moderate Balance – Approaching standard service ratios
> 2.0 and ≤ 3.5	2–3.5 kindergartens per 1,000 children	Adequate Supply – Best-performing districts (e.g., Sai Kung, NE Lantau)
> 3.5	> 3.5 kindergartens per 1,000 children	Oversupply – Potential for consolidation or resource reallocation

Table 9: Demand-Adjusted Allocation Deviation (ADI) Framework

Variable	Symbol	Type	Formula / Definition
Observed KD	KD	Density	Actual kindergarten density = Join_Count / area_sqkm
Predicted KD	$predicted_KD$	Density	Model output from regression: $KD_i = \beta_0 + \beta_1(Child_Density_i) + \epsilon_i$ where $\beta_0 = 0.0638$, $\beta_1 = 0.007662$, $R^2=0.7554$
Accessibility Difference Index	ADI	Residual	$ADI = KD - predicted_KD > 0$: Oversupply relative to demand < 0 : Undersupply relative to demand
Standardized Residual	z	Statistical	Residual / Standard Deviation of residuals (σ)
Supply_Status	—	Categorical	Classification based on standardized residuals: • Over-Supply: $> +2.0\sigma$ • Potential Imbalance: $\pm 1.5\sigma$ to $\pm 2.0\sigma$ • Balanced: 1.5σ to $+1.5\sigma$ • Under-Supply: $< 2.0\sigma$
Supply_Demand_Ratio	—	Ratio	$(\text{Kindergarten_Count} / \text{Children_3.5}) \times 100$ (Hong Kong average = 0.80)

Table 10: Local Indicators of Spatial Association (LISA) Variables

Variable	Abbr.	Definition / Values
Local Moran’s I	<i>LMI</i>	Local spatial autocorrelation statistic measuring correlation between ADI at location i and its neighbors
z-score	<i>LMZ</i>	Standardized test statistic for LMI under spatial randomness null hypothesis
p-value	<i>LMP</i>	Statistical significance of local cluster
Significance Flag	<i>Sig05</i>	1 if $p < 0.05$, otherwise 0
Spatial Lag of ADI	—	Weighted average of ADI values in neighboring STPUGs (spatially weighted)
Cluster/Outlier Type	<i>COType</i>	High-High: High ADI surrounded by high ADI (oversupply cluster) Low-High: Low ADI surrounded by high ADI (undersupply pocket in oversupplied area) High-Low: High ADI surrounded by low ADI (oversupply pocket in undersupplied area) Not Significant: No significant spatial autocorrelation

Table 11: Intervention Thresholds and Operational Definitions

Parameter	Intervention Trigger / Spatial Outcome Target
Expansion Site	Positive residual exceeding 1.5 units indicates supply significantly exceeds demand; investigate for reallocation to other sites or consolidation from this site if $> 2.0\sigma$
Consolidation Site	Negative residual exceeding magnitude indicates severe undersupply; priority for new construction
Critical Shortage	$> 2,000$ children per kindergarten
Service Gap	No_Access = 1
Spatial Efficiency Target	Reduction in significant LISA clusters from 7 to $\leq 5STPUGs$